DIVIDED POWER STRUCTURES ON TORSIONFREE RINGS: THE N!-CRITERION

THE STACKS PROJECT AUTHORS

Editorial note. The Q-case verifications are flagged 'straightforward to verify ... We encourage the reader to do the verifications to make sure that we have the coefficients correct'; axiom (4) is displayed via the binomial formula and the reduction to Q-algebras is executed in-span. Remaining checks are mechanical rational arithmetic. Cross-chapter same-name note: more-morphisms.tex has an unrelated lemma-silly (transcendental element prime avoidance); conclusions differ entirely, counted separately.

Allowed standard prerequisites (editorial).

- Definition of a divided power structure. definition-divided-powers (same chapter, definitional anchor). Supplies axioms (1)-(5) used throughout all four parts.

Source reference. Original definition 07GL: sources/stacks-dpa/pinned-original-source.tex, source lines 253–253. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let A be a ring. Let I be an ideal of A.*

(1) *If γ is a divided power structure*[1] *on I, then $n!\gamma_n(x) = x^n$ for $n \geq 1$, $x \in I$.*

Assume A is torsion free as a $\mathbf{Z}$-module.

(2) *A divided power structure on I, if it exists, is unique.*

(3) *If $\gamma_n : I \to I$ are maps then*

$$\gamma \text{ is a divided power structure} \Leftrightarrow n!\gamma_n(x) = x^n \ \forall x \in I, n \geq 1.$$

(4) *The ideal I has a divided power structure if and only if there exists a set of generators x_i of I as an ideal such that for all $n \geq 1$ we have $x_i^n \in (n!)I$.*

Proof. Proof of (1). If γ is a divided power structure, then condition (2) (applied to 1 and $n-1$ instead of n and m) implies that $n\gamma_n(x) = \gamma_1(x)\gamma_{n-1}(x)$. Hence by induction and condition (1) we get $n!\gamma_n(x) = x^n$.

Assume A is torsion free as a $\mathbf{Z}$-module. Proof of (2). This is clear from (1).

Proof of (3). Assume that $n!\gamma_n(x) = x^n$ for all $x \in I$ and $n \geq 1$. Since $A \subset A \otimes_{\mathbf{Z}} \mathbf{Q}$ it suffices to prove the axioms (1) – (5) of Definition 07GL in case A is a $\mathbf{Q}$-algebra.

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[1]Here and in the following, γ stands short for a sequence of maps $\gamma_1, \gamma_2, \gamma_3, \ldots$ from I to I.

In this case $\gamma_n(x) = x^n/n!$ and it is straightforward to verify (1) – (5); for example, (4) corresponds to the binomial formula

$$(x+y)^n = \sum_{i=0,\ldots,n} \frac{n!}{i!(n-i)!} x^i y^{n-i}$$

We encourage the reader to do the verifications to make sure that we have the coefficients correct.

Proof of (4). Assume we have generators x_i of I as an ideal such that $x_i^n \in (n!)I$ for all $n \geq 1$. We claim that for all $x \in I$ we have $x^n \in (n!)I$. If the claim holds then we can set $\gamma_n(x) = x^n/n!$ which is a divided power structure by (3). To prove the claim we note that it holds for $x = ax_i$. Hence we see that the claim holds for a set of generators of I as an abelian group. By induction on the length of an expression in terms of these, it suffices to prove the claim for $x + y$ if it holds for x and y. This follows immediately from the binomial theorem. $\square$